

CROST: a comprehensive repository of spatial transcriptomics

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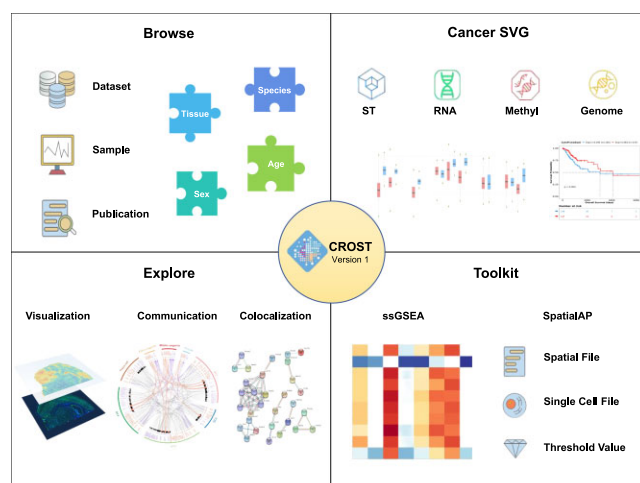
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Abstract

The development of spatial transcriptome sequencing technology has revolutionized our comprehension of complex tissues and propelled life and health sciences into an era of spatial omics. However, the current availability of databases for accessing and analyzing spatial transcriptomic data is limited. In response, we have established CROST (<https://ngdc.cncb.ac.cn/crost>), a comprehensive repository of spatial transcriptomics. CROST encompasses high-quality samples and houses 182 spatial transcriptomic datasets from diverse species, organs, and diseases, comprising 1033 sub-datasets and 48 043 tumor-related spatially variable genes (SVGs). Additionally, it encompasses a standardized spatial transcriptome data processing pipeline, integrates single-cell RNA sequencing deconvolution spatial transcriptomics data, and evaluates correlation, colocalization, intercellular communication, and biological function annotation analyses. Moreover, CROST integrates the transcriptome, epigenome, and genome to explore tumor-associated SVGs and provides a comprehensive understanding of their roles in cancer progression and prognosis. Furthermore, CROST provides two online tools, single-sample gene set enrichment analysis and SpatialAP, for users to annotate and analyze the uploaded spatial transcriptomics data. The user-friendly interface of CROST facilitates browsing, searching, analyzing, visualizing, and downloading desired information. Collectively, CROST offers fresh and comprehensive insights into tissue structure and a foundation for understanding multiple biological mechanisms in diseases, particularly in tumor tissues.

Graphical abstract



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Introduction

Spatial transcriptomics is an emerging field that focuses on understanding gene expression patterns in the context of tissue spatial organization (1,2). While traditional and single-cell transcriptomic techniques offer valuable insights into gene expression levels at tissue and cellular levels, they lack information on the spatial distribution of gene expression within tissues (3,4). Spatial transcriptomics, on the other hand, combines spatial information with transcriptomic data to provide a comprehensive understanding of tissue structure and mechanistic insights into various biological processes, including tissue development (5,6) and disease (7–9). Currently, over 50 spatial transcriptomic technologies, predominantly classified as sequencing-based and imaging-based methods, have been developed (10).

Sequencing-based technologies utilize spatial barcodes to capture *in situ* transcripts before conducting next-generation sequencing (NGS), followed by the analysis of barcode sequences to map their original positions (11). Most spatial transcriptomics research is based on 10X Visium (12), one of the most popular sequencing-based technologies (10). However, neither 10X Visium nor similar technologies, such as Slide-seq V2 (13), have achieved single-cell resolution (14). Conversely, imaging-based technologies employ fluorescent probes to detect target sequences within tissue samples. Representative imaging-based technologies include 10X Xenium (15), MERFISH (16) and NanoString COX (17).

The development of spatial transcriptomic technology has significantly impacted the field of biomedicine, finding applications in tissue development (18,19), tissue homeostasis (20,21), the disease microenvironment (22), and the tumor microenvironment (23,24). For instance, Su *et al.* utilized a high-resolution spatial transcriptome of colorectal cancer to reveal an interaction between FAP + fibroblasts and SPP1 + macrophages, suggesting potential therapeutic strategies for colorectal cancer (25). Wang *et al.* mapped transcriptional dynamics and directionality during mouse organogenesis with single-cell resolution and high sensitivity, shedding light on spatial cell heterogeneity and cell fate in developing tissues, such as the brain (6).

The recent surge in spatial transcriptomic data calls for a user-friendly database system that facilitates easy access to data, visualization, and personalized analysis. Several databases, including SpatialDB (26), Aquila (27), SPAScer (28), SODB and STomicsDB, have been developed to maintain spatial transcriptomic data. Although they made valuable efforts for spatial transcriptomics, these databases have some significant limitations and challenges. For instance, SpatialDB includes only 24 transcriptomic datasets and lacks spatial analysis. In parallel, both SODB and STomicsDB provide interactive visualization platforms, but regrettably, they also lack spatial analysis. Aquila lacks tissue image information and does not provide integrated analysis with single-cell transcriptome data, thereby restricting the ability of users to gain a deeper understanding of the datasets. Similarly, SPAScer lacks interactive spatial visualization tools and online spatial data analysis platforms, rendering it less suitable for most biologists. Moreover, both Aquila and SPAScer perform downstream analyses but do not effectively integrate the results to explore biological processes comprehensively, hindering users from gaining deeper insights into spatial transcriptomics data.

To address these limitations and provide a comprehensive solution, we developed a comprehensive repository of spatial transcriptomics, CROST, which contains 1033 spatial transcriptome samples from eight species, 35 tissues, and 56 diseases, each containing expression information and spatial location details (image files). The aim of CROST was to create a comprehensive repository with various advantages, including an advanced analysis pipeline, interactive visualization tools, correlation and colocalization analyses, integration of omics data and a user-friendly platform for personalized analysis in spatial transcriptomics. Collectively, our novel pipeline was designed to offer a wide range of features and enhancements to meet the needs of the scientific community.

Materials and methods

Data collection

Spatial transcriptomics datasets were acquired and downloaded from databases such as the National Center for Biotechnology Information (NCBI) (29), European Bioinformatics Institute (EBI) (30), China National Center for Bioinformatics (CNCB) (31), Broad Institute Single Cell Portal (32), 10× Genomics (12) and DNA Data Bank of Japan (DDBJ) (33). Keyword searches were conducted using terms such as ‘spatial transcriptomics’, ‘spatial transcriptome’, ‘spatial genome’, ‘spatial RNA-seq’ and ‘spatial sequencing.’ Furthermore, additional searches were conducted on PubMed and bioRxiv to enhance the datasets, focusing on relevant literature (the query was performed before June 2023). The collected datasets underwent screening based on the following criteria: (i) availability of original sequencing files and (ii) presence of spatial location information or image files. After a thorough screening, 182 spatial transcriptome datasets, encompassing 1033 samples derived from eight different species, 35 tissues and 56 diseases, were included.

Preprocessing

Quality control and filtering procedures were executed on the raw sequencing data to remove low-quality and duplicate reads. The Spliced Transcripts Alignment to a Reference (STAR) (34) was used to align the sequencing reads to the reference genome of each species, enabling the identification of expressed genes and their corresponding expression levels in individual cells. Subsequently, the transcriptome positions of individual cells were mapped onto spatial coordinates. Harmony (35) was employed to remove batch effects when multiple samples were present in the datasets.

Dimensionality reduction and clustering

For dimensionality reduction and clustering analysis on spatial data, the Bayesian-based Space-clustering for spatial transcriptomics (BayesSpace) tool (36) was utilized. The data were initially preprocessed using the spatialPreprocess, and the optimal number of clusters for dividing the spatial region was determined using qTune. Gene Set Variation Analysis (GSVA) was used to calculate single-sample gene set enrichment analysis (ssGSEA) scores for each cluster (37), while considering the cell state (38) and biological process (39).

Spatially variable gene analysis

The SPARK tool (40) was used to identify spatially variable genes (SVGs), and the calculation of *P*-values was performed using the Satterthwaite method. The significance threshold of adjusted *P*-value was set at <0.01. Additionally, Gene Ontology (GO) (41) and Kyoto Encyclopedia of Genes and Genomes (KEGG) (42) enrichment analyses of the identified genes were performed using the clusterProfiler tool (43).

Cell type annotation analysis

Datasets from single-cell RNA sequencing (scRNA-seq) were incorporated to achieve spatial transcriptome data with single-cell resolution. The raw counts of the single-cell data were normalized using SCTransform in Seurat (44). Subsequently, principal component analysis (PCA), dimensionality reduction, and clustering were conducted using RunPCA, FindClusters and RunUMAP, respectively. Analysis of differentially expressed genes within each cluster was performed using FindAllMarkers. Finally, the cell type annotation on the spatial transcriptome data was conducted using the Spatially-resolved Transcriptomics cell type identification tool (SPOTlight) (45) to determine the proportion of cell type composition for each spot.

Spatial correlation and colocalization analysis

Spatial correlation and colocalization analysis help elucidate the regulatory networks and biological processes within cells, providing valuable insights into the functions of both individual cells and tissues. Based on the cell type annotation results, correlation analysis was performed using the plotCorrelation-Matrix function from SPOTlight, while colocalization analysis was conducted with plotInteractions function, also provided by SPOTlight. The results were visually represented as a heat map.

Communication analysis

Distinct spatial structures and arrangements of cell types within tissues and organs often lead to more frequent contacts and interactions among neighboring cells (46). Analyzing the interactions among spatially adjacent cell types provides valuable insights into communication networks and signaling mechanisms between cells. To accomplish this, we employed CellChat (47), leveraging the clustering and cell type annotation results, which was used to analyze communication patterns and mechanisms between tissue regions and neighboring cell types.

Database framework and web implementation

The CROST web user interfaces were built using Vue3.js (<https://vuejs.org/>) and Vite (<https://vitejs.dev/>). The back-end API request was developed using Node.js (<https://nodejs.org/en/>) and deployed in a CentOS Linux environment. The back-end database was powered by MongoDB (<https://www.mongodb.com/>), a widely used document-based data model. Interactive visualization charts were provided through Echarts (<https://echarts.apache.org/en/index.html>), and CirroCumulus (48) was used to provide interactive visualization charts. Furthermore, the basic data analysis and processing were implemented using R 4.1 (<https://www.r-project.org/>).

Database contents and usage

Overview of CROST

Figure 1 illustrates the design and construction of CROST, which provides 182 high-quality spatial transcriptome projects covering 1033 sub-datasets across eight different species (*Homo sapiens*, *Mus musculus*, *Danio rerio*, *Gallus gallus*, *Canis lupus familiaris*, *Phalaenopsis aphrodite* and *Rattus norvegicus*), 35 tissue types, and 56 diseases. Each dataset in CROST includes valuable expression data and precise spatial location information in the form of image files. This comprehensive integration of expression and spatial data empowers researchers to thoroughly investigate and analyze the datasets from a spatial perspective. CROST comprises four main modules, namely browse, cancer SVG, explore and on-line analysis modules (Figure 1).

Browse module

The browse module enables efficient navigation through the extensive collection of datasets, providing convenient access to meta-information, literature resources, and single-sample analysis results. Within the single-sample analysis section, 168 cell types were annotated, with endothelial cells, lymphocytes, epithelial cells, fibroblasts and macrophages being shared across multiple tissues. Furthermore, this module provides users access to the spatial transcriptomic project, samples and literature for browsing and downloading (<https://ngdc.cncb.ac.cn/crost/browse/>).

CROST adopts a structured management model that integrates comprehensive project and sample metadata, thereby enhancing data presentation, exploration, and visualization. Every project within CROST summarizes 22 metadata items, utilizing a controlled vocabulary that encompasses information on the data source, overall design, tissue/cell line, disease state, and sample type (Figure 2A). Moreover, CROST offers 38 structural items to standardize sample metadata, including basic information, sample characteristics, biological status, experimental protocols, and quality assessment (Figures 2A, B). The structured metadata is conveniently presented in a tabular format, allowing users to effortlessly comprehend and utilize the collected samples (Figure 2C).

In each project, the analysis pipeline in CROST is built upon raw sequencing data of each sample, providing seven aspects of spatial transcriptomics data illustration: data overview, dimensionality reduction and clustering (Figures 2D–F), SVGs, cell type annotation (Figure 2G), cell type correlation and colocalization (Figures 2H, I), cluster–cluster communication analysis (Figure 2J), and cell–cell communication analysis (Figure 2K). The statistics of count, gene, mitochondrion and ribosome are displayed as violin and spatial plots for data overview. Cluster analysis is used to identify tissue regions according to the optimal number of clusters. As spatial clusters represent tissue regions with distinct functions, the single-sample gene set enrichment score is calculated for each cluster. Cell states (38) and biological processes (39) are considered. SVG analysis is conducted, followed by GO (41) and KEGG (42) enrichment analyses, are conducted to characterize the biological functions and processes of the identified variant genes. To achieve spatial transcriptome data at single-cell resolution, spatial transcriptome samples are matched with single-cell transcriptome data, and the SPOTlight tool (45) is used for cell type annotation. Subsequently, cell type correlation and colocalization analyses are performed to understand spatial

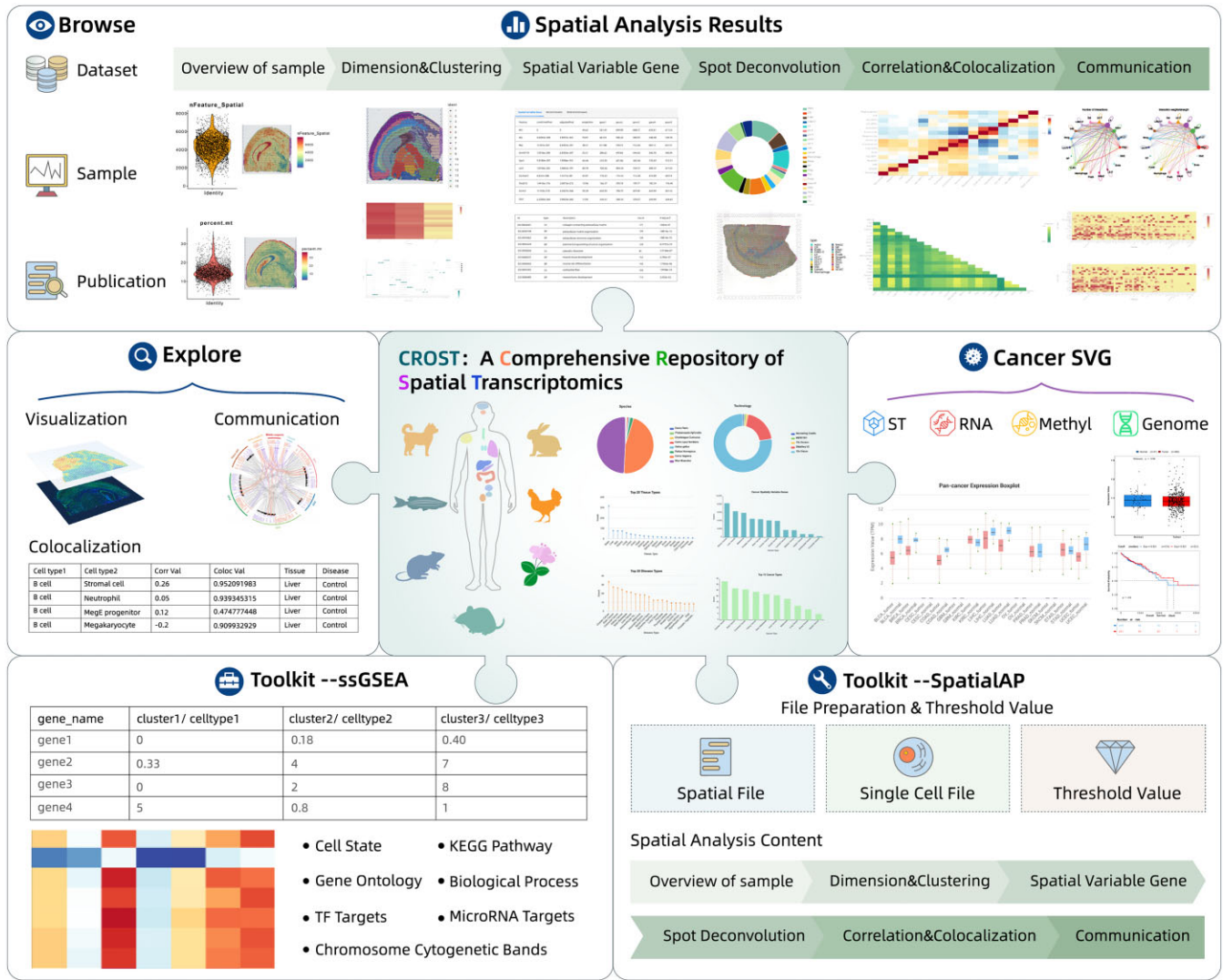


Figure 1. Schematic overview of CROST. It mainly includes four modules: Browse, Explore, Cancer SVG and Toolkit.

gene expression patterns and tissue organization, as well as elucidate gene regulatory networks and biological functions. Additionally, cluster-cluster interaction analysis and cell-cell interaction analysis are performed to explore interactions between tissue regions and cell types, as well as to decipher communication mechanisms.

Cancer SVG module

The Cancer SVG module displays the sources and characteristics of 48043 SVGs significantly enriched primarily in the kidney cancer (8323 genes), liver cancer (6380 genes) and melanoma (5964 genes) (<https://ngdc.cncb.ac.cn/crost/cancer-svg>). Notably, this module combines data from spatial transcriptome, classical transcriptome, epigenome, and genome to comprehensively elucidate the SVGs. Users can search for genes and cancer types of interest in this module. The Gene Summary page displays detailed information about the gene, including basic information, gene expression *in situ*, gene expression, DNA methylation, copy number variation, survival analysis, and related literature information. For each SVG, the quantitative comparison among cancer types, between normal and tumor tissues, and its relationship with prognosis was si-

multaneously calculated and illustrated from the level of gene expression, DNA methylation and genomic CNV. Collectively, this module helps obtain a better understanding of the role of these genes in tumor progression and prognosis.

The Cancer SVG module serves as a valuable tool for users, particularly clinicians, to quickly assess the expression level, methylation level, and copy number variation of a specific gene in a particular cancer type. Additionally, it helps determine the correlation between the gene and its clinical prognosis, aiding in identifying candidate genes for prognostic prediction panels and potentially facilitating targeted therapies in the future.

We conducted a case study to showcase the exceptional performance of the module. Using the cancer SVG module, the apoptosis antagonizing transcription factor (*AATF*) was identified as an SVG in liver cancer. Our analysis revealed that *AATF* exhibited elevated expression levels (Figures 3A, B) in HCC and was significantly correlated with a poorer prognosis (Figure 3C) in HCC patients. These findings align with those by previous studies (49,50), further validating the effectiveness of the module in identifying SVGs and their potential implications in cancer prognosis.



Figure 2. Screenshot of the page for projects, samples and single sample analysis results in CROST. **(A)** Multiple filter items in the browse page for projects. **(B)** Structural descriptive information in the browse page for samples. **(C)** Manually curated meta-information in the browse page for samples. **(D)** Spatial clustering analysis. **(E)** Single sample enrichment analysis of cell state related gene set. **(F)** Single sample enrichment analysis of biological state and process related gene set. **(G)** Cell type deconvolution. **(H)** Spatial correlation analysis. **(I)** Spatial colocalization analysis. **(J)** Cluster-cluster communication analysis. **(K)** Cell-cell communication analysis

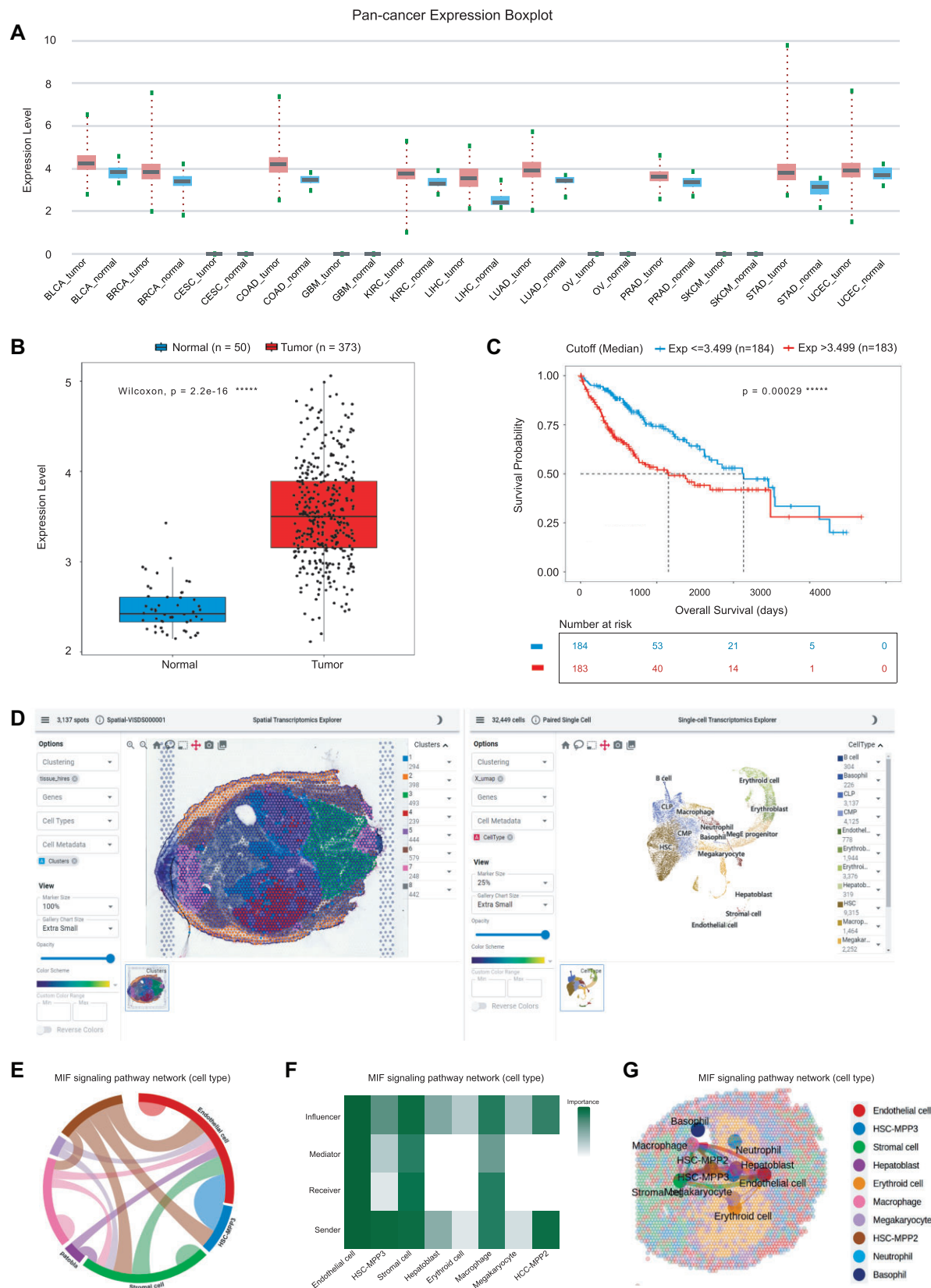


Figure 3. Screenshot of the pages for the cancer SVG module (using the transcriptome as an example) and the explore module in CROST. **(A)** Expression levels of spatially variant genes in pan-cancer. **(B)** Expression levels of spatially variant genes in specific cancer. **(C)** Survival analysis of specific cancer patients based on the expression level of spatially variant genes. **(D)** Interactive visualization of spatial transcriptomics samples and paired single-cell transcriptomics samples. **(E)** Chord diagram of intercellular communication network for specific signaling pathway. **(F)** Heatmap of signaling roles (senders, receivers, mediators and influencers) for specific signaling pathway. **(G)** Spatial images of intercellular communication networks for specific signaling pathway.

Explore module

The Explore module (<https://ngdc.cncb.ac.cn/crost/analyze/spatial-explorer>) offers an interactive environment for visualization, communication, cell-type colocalization, and cell-type correlation. The spatial visualization component of CROST utilizes CirroCumulus applications (48) to provide interactive visualization of spatial transcriptome data. It offers Uniform Manifold Approximation and Projection (UMAP) and spatial slice information for each sample. The program has been improved to display the proportions of cell types in spatial spots, overcoming the existing drawback. CROST allows users to simultaneously view spatial transcriptome data alongside paired single-cell transcriptome data, providing a comprehensive view of gene expression and cell type distribution at both the single-cell and spatial transcriptome levels (Figure 3D). Recognizing the importance of cell-cell interactions in disease progression, CROST includes an online spatial communication exploration page. Users can specify samples and signaling pathways of interest to identify the cell populations and their communications involved in this pathway (Figure 3E). Moreover, the roles of signaling molecules, such as primary senders and receivers, in specific cell populations can be investigated (Figure 3F). The spatial distribution of these cell populations can also be visualized (Figure 3G).

In addition to spatial visualization and communication exploration, CROST integrates the results of correlation and colocalization analyses on a spatial colocalization page. This integration enables users to search and retrieve relevant analysis results using keywords such as cell type, tissue, species, and disease. Notably, we observed a robust correlation and significant spatial overlap between FAP+ fibroblasts and macrophages in all colorectal cancer (CRC) samples analyzed, which aligns with the results by Su *et al.* (25).

Online analysis module

The online analysis module empowers users to conduct personalized analyses of spatial data without requiring programming skills.

ssGSEA

The ssGSEA tool in CROST (<https://ngdc.cncb.ac.cn/crost/tools/ssgsea>) offers the capability to input raw count matrices for a specific sample. By leveraging the ssGSEA algorithm, this tool estimates the relative enrichment of 7 distinct gene sets within the given sample (37). These gene sets cover various aspects, including cancer biological processes, cell states, chromosome cytogenetic bands, gene ontology, KEGG pathways, MicroRNA targets, and transcription factor targets (38,39). Additionally, CROST provides users with the option to package and download all ssGSEA results, ensuring ease of access for further analysis. This feature empowers researchers to retrieve and utilize the analysis outcomes for downstream analysis or integration with other tools and platforms.

SpatialAP

SpatialAP (<https://ngdc.cncb.ac.cn/crost/tools/spatial-pipeline>) is a one-stop analysis platform designed for spatial transcriptome analyses, aiming to address the fundamental requirements of spatial transcriptome analysis. The platform offers a wide range of analysis capabilities, including quality control, dimensionality reduction clustering, SVG analysis,

cell type annotation (by integrating single-cell transcriptome data), colocalization analysis, correlation analysis, cell communication analysis, and biological function enrichment analysis.

To accommodate diverse user needs, SpatialAP provides three options for analysis:

1. upload single-cell transcriptome data and select spatial transcriptome data from the CROST database for analysis.
2. upload spatial transcriptome data and select single-cell transcriptome data from the CROST database for analysis.
3. simultaneously upload spatial and single-cell transcriptome data for analysis.

Upon processing the uploaded files, SpatialAP promptly notifies users via email, allowing them to access the analysis results through the CROST platform. CROST processes the data locally and securely stores the results to safeguard sensitive private information in patient samples or unpublished research data. This ensures data security, privacy, and compliance with relevant ethical and legal requirements.

Concluding remarks and future development

The emergence of spatial transcriptome technology has led to a substantial transformation in biomedicine (51,52). As a distinctive and comprehensive resource, CROST plays a pivotal role in integrating spatial transcriptome data from various species, tissues, and diseases. It offers researchers a multitude of valuable features, including a repository of high-quality samples, standardized analysis pipelines, interactive visualization tools, correlation and colocalization analyses capabilities, seamless integration of multi-omics data, and a user-friendly analysis platform. As an integral part of the National Genome Data Center's resources (31), CROST is poised to evolve into an open-access, all-in-one platform catering to spatial transcriptomics research.

With the continuous advancements in sequencing technologies, high-resolution spatial multi-omics datasets, including metabolomics and methylomics, are becoming increasingly available. In the future, we plan to incorporate these efficient spatial multi-omics datasets into CROST to further explore interactions and co-regulations across different omics levels, thereby contributing to a deeper comprehension of molecular mechanisms and tissue functions in organisms. Moreover, CROST will enhance data integration efforts by seamlessly incorporating diverse datasets and establishing connections with other omics databases.

In conclusion, through its comprehensive spatial transcriptome analysis capabilities, CROST will provide researchers with profound insights into tissue structure, organ development, and disease occurrences. It is poised to drive substantial breakthroughs in the field of spatial transcriptomics. With its continued growth and incorporation of emerging technologies, CROST is well-positioned to maintain its important role in spatial transcriptomics research, facilitating researchers in unraveling the complexities of spatial gene expression patterns and advancing our understanding of biological systems.

Data availability

CROST is freely available online at <https://ngdc.cnbc.ac.cn/crost>.

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Conflict of interest statement

None declared.

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